

Research Plan - Gunnar Felix Lange

Overview: From quantum topology to geometry, sensing and control

My research seeks to understand how the collective quantum behavior of $\sim 10^{23}$ particles can give rise to remarkably robust and simple quantum phenomena and how these phenomena can be exploited. I approach this question through four connected themes: **quantum topology, geometry, sensing and control**.

These themes form a natural progression. **Quantum Topology** describes **robust global properties** of quantum states. **Quantum geometry** characterizes how these states **vary locally** and therefore how they respond to external perturbations. **Quantum sensing** asks how this sensitivity can be turned into useful information. Finally **Quantum control** asks how this information can be extracted. My long-term goal is to connect these themes, with research spanning from foundational theory building, through realistic materials simulations and experimental collaborations, to emulating real quantum technologies.

Research background: topology in theory and materials

My research background, developed during my PhD at Cambridge and my first postdoctoral position at the Max-Planck Institute for the Physics of Complex Systems in Dresden, is centred on **topological quantum materials**. What particularly attracts me to this field is that it spans from research-level mathematics to experimental material physics. I am interested in understanding quantum systems **not only** for their fundamental **beauty**, but also for their **observable and useful consequences**. My work has therefore focused both on **classifying new topological phases** and on **characterizing their signatures in real materials**.

For theoretical classification, I use simplified tight-binding models to **isolate robust** features of complicated quantum states. Highlights of this work include:

- **Discovering subdimensional topologies** [1,2], a new topological phase displaying unusual surface spectra including coexisting gapped and gapless surface states;
- **Classifying a new topological protection mechanism** in phonons, present in all two-dimensional materials and providing a new perspective on the Nambu-Goldstone theorem [3];
- **Exploring** the interplay between topology in **energy and spin eigenstates** [4], with the surprising result that spin and energy topology can be inversely related in certain (fragile) topological systems.

To connect topological classification to real materials, I employ **density-functional theory (DFT)** and **Wannierization**, often in direct collaboration with experimental groups. Work in this area includes **identifying** an experimentally discovered weak topological state in RhBi_2 [5], **predicting** the interplay of chirality and topology in phonons of twinned quartz [6], and **demonstrating** that projected spin topology can occur in monolayer Bi and be tuned using light [7]. Equally importantly, I have also shown that topology is **not** responsible for several experimentally observed phenomena, including photoinduced helicity-dependent currents in Mn_3Sn [8] and unusual surface states in FeSb_2 [9].

These results have inspired me to move beyond the **classification and characterization** of topological phases to a deeper question: **what physical consequences does topology have for a quantum system, and how does it interact with other properties of the quantum state?** This question has led me to initiate an independent research programme investigating the **physical consequences of topology**, particularly its relationship with **quantum geometry, quantum sensing and quantum control**. I have received an **MSCA co-funded postdoctoral fellowship** and a **YoungCAS project** to pursue these connections.

Research direction 1: From topology to quantum geometry

Quantum geometry is characterized by the **quantum metric**, which measures how rapidly a quantum state changes **locally** across a parameter space. It is therefore **complementary** to topology, which describes **global** winding. The local variations are physically important, as quantum geometry enters a **broad range** of material responses, including optical, superconducting and strain-related phenomena, and has recently become a major research area [10].

1.1: Decomposing quantum geometry into topological contributions

The connection between the quantum metric and topology therefore provides a **novel way** of translating abstract topological classifications into physical responses. Recent work by Liang Fu's group at MIT showed

that topology places **rigorous lower bounds** on the quantum metric [11,12]. Intuitively, a quantum state that winds globally cannot remain locally unchanged everywhere. A topologically non-trivial state is therefore **guaranteed** to possess some minimum amount of quantum geometry.

I recently **initiated and led** a study [13] showing that this principle extends also **beyond energy eigenstates**, showing that **spin topology also bounds quantum geometry**. This demonstrates that the quantum metric can encode topology associated with different operators, raising the possibility of **diagnosing** topology through geometric observables. Liang Fu's group has very recently built on our work, generalizing these relations further to topology associated with arbitrary symmetry operators [14].

This result naturally raises the question: **if topology constrains quantum geometry, how much of the geometry is actually topological?**

Work with one of my students has shown that the answer can be surprisingly subtle. In [15], currently under review in PRX, we demonstrated that a substantial part of the geometry in tight-binding models can arise simply from the **orbital embedding**, independently of topology.

The combination of these two results defines one of the major directions I want to pursue going forward: **disentangling different physical origins of quantum geometry**. To address this, I will study decompositions of occupied states into **general operator-defined subspaces**, extending the framework of [13,14]. This will provide a systematic way of distinguishing local and global contributions to quantum-state variation and identifying which components of quantum geometry are genuinely constrained by topology.

1.2: Geometry and entanglement

This perspective also connects directly to materials. Decomposing occupied states with respect to spin, for example, can quantify spin-orbit coupling along different spatial directions, as we suggested in [7], and may provide a useful framework for understanding **spin-orbital entanglement**. Crucially, this decomposition also leads to questions about the more general relationship between **geometry and entanglement**. The role of entanglement is poorly understood in solid-state materials. It is well known, however, that the quantum metric provides **information on entanglement**, as it is closely related to the **Quantum Fisher Information** (QFI), which is an important entanglement witness [16]. I am currently developing an application to the Norwegian Research Council's (NRC) Young Research Talent programme around this connection. My goal is to understand how **topology and geometry** can **shed new light** on **entanglement** in materials.

1.3: Robustness in real materials

A closely related question is **how robust these effects are in real materials**. I have already taken a first step toward this in [13], where I showed that spin topology and geometry bounds **survive moderate onsite disorder** in ultrathin Bi. I plan to extend this to more realistic disorder descriptions and additional material systems, combining DFT-based modelling with experimental input. This is important to understand which geometric and topological effects are useful in a quantum technology context. As a first step, I am currently seeking to recruit an MSc student, together with Assoc. Prof. Bathen at UiO, on this topic.

1.4: Development of new community tools

Understanding topology and geometry in materials also requires better computational tools, which I am already developing and plan to continue to develop.

First, I am leading the development of the open-source tool **AutoEuler** [17], which automatically computes Euler topology. This topology is characterized by a non-Abelian structure, making it particularly interesting but also unusually difficult to determine systematically. AutoEuler has already been applied to a database of phononic systems [18], uncovering new braiding topologies. Together with Dr. Peng at Cambridge and Asst. Prof. Bennett at Nanyang Technological University, I plan to extend this to electronic systems, providing the first systematic high-throughput classification of this form of non-Abelian topology in electronic systems.

Second, our work on orbital embedding [15] has unexpectedly suggested a way to make simple models which **quantitatively predict geometric responses**. Conventional tight-binding models are designed primarily to reproduce band energies and can perform poorly for responses that depend sensitively on wavefunctions. We have shown, however, that **optimizing the orbital embedding alone** can allow tight-binding models of GaAs and CdS to quantitatively reproduce **nonlinear optical responses**. I plan to develop this methodology of **optimal orbital embeddings** more systematically, and ultimately incorporate it into widely used packages such as Wannier90 or WannierTools.

Research direction 2: From quantum geometry to quantum sensing

Once quantum geometry is viewed as a measure of how strongly a quantum state changes under an external perturbation, it becomes clear that **quantum geometry is closely related to quantum sensing**. The link is provided by the aforementioned QFI. The QFI quantifies how distinguishable nearby quantum states become when a parameter is changed and determines the best possible precision with which that parameter can, in principle, be estimated. The quantum metric captures **precisely** this local variation of the state.

This gives the topology–geometry connection a new interpretation. If topology places a lower bound on quantum geometry, it can also guarantee a minimum sensitivity to certain perturbations. Topology may therefore provide not only a classification of robust winding, but also a **resource for quantum sensing**.

2.1: Measuring the QFI

The first challenge to address is how to measure the QFI itself. In a recent **single-author preprint** [19], I proposed a scheme for measuring the QFI of an arbitrary two-band system, with respect to any parameter, using controlled changes of polarization and incidence angle in **photoemission spectroscopy**. The method allows the quantum metric and QFI to be reconstructed, as I demonstrate in graphene and hBP.

I plan to extend this method to multiband systems and pursue experimental implementations. Conceptually, the project points toward a broader objective of **measuring** quantum geometry and sensing capability.

2.2: Critical topological quantum sensors

The next question is whether topological and geometric materials can act as useful sensors. One particularly promising possibility in this direction arises in **topological phase transitions**. Near such a phase transition, the quantum metric can become **very large**, making the state **extremely sensitive** to the parameter driving the transition, which is the idea behind **critical quantum sensing** [20,21]. I am already investigating this possibility for strain sensitivity in group-V materials exhibiting spin topology, together with Dr. Peng, and in non-Hermitian topological phases together with Prof. Bergli and a jointly supervised MSc student at UiO.

Conventional critical sensors, however, face an important limitation: closing the energy gap also increases the **timescale required for adiabatic operation** [20]. Spin topology suggests a possible way around this trade-off. A **spin gap may close without the energy gap closing**, potentially allowing strongly enhanced sensitivity **without the same adiabatic slowdown**. I plan to investigate whether this can provide a genuine sensing advantage, both for spin topology and for higher-form topology, together with Prof. Bzdušek at the University of Zurich and Dr. Jankowski at the Max-Planck Institute for the Physics of Complex Systems.

This is the longer-term scientific destination of the topology–geometry–sensing connection: identifying cases in which this relationship leads to a **measurable and operational advantage**.

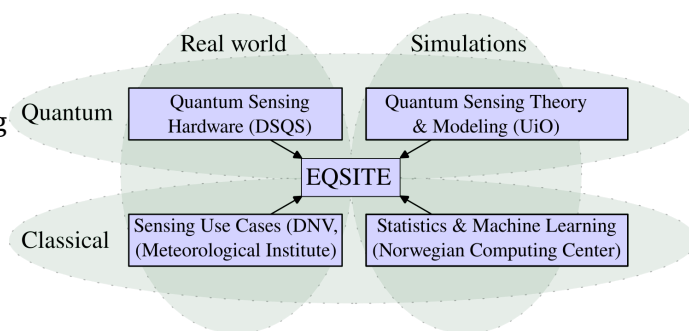
Research direction 3: From quantum sensing to quantum control

A large QFI, however, still does not guarantee a useful sensor. The quantum state of the sensor must also be initialized, controlled and measured, unwanted noise must be separated from the signal, and any theoretical enhancement must survive with finite experimental resources. The progression from **topology to geometry to sensing** therefore naturally leads to a **quantum control problem**: how can optimal sensitivity be realized in actual hardware? This connects closely with UiO's proposed Centre of Excellence in quantum control.

3.1: Emulating realistic quantum sensors

I am already working on this question in a collaboration on **electrons on liquid helium** with the American quantum-technology company Eeroq [22]. We have shown how to optimize the generation of entanglement between two electrons in a real device, and demonstrated that this entanglement enhances the QFI for differential magnetic-field sensing. I plan to extend this work toward systems with more electrons and additional fields, allowing us to ask how sensing performance scales when realistic operations and device constraints are included.

The same philosophy underpins **EQSITE – Emulating Quantum Sensors for IndusTrial Experimentation**, a project I recently submitted as main PI to the NRC call on breakthrough enabling technologies. EQSITE proposes to develop emulators for several sensor architectures,



including point defects in silicon carbide, ultracold atoms and electrons on helium. The emulator will incorporate realistic noise together with experimentally available initialization, control and readout operations and asks: **what is the best sensing performance this device can actually achieve?** The project consortium, shown above, comprises the new national quantum centre **DSQS** at UiO, the **Norwegian Computing center** and potential end users including **DNV** and the **Norwegian Meteorological Institute**, with me as the **principal investigator**. As a step towards this project, I am supervising an MSc student who is studying optimal sensing protocols under realistic noise using data supplied by DNV.

3.2: Finding niches for quantum sensors

My broader goal is to close the gap between quantum-metrology theory and technological adoption. Moving beyond idealized assumptions, I want to determine **under which realistic conditions quantum advantage remains useful**. This perspective also motivates my work on studying other sensing environments. Together with Dr. Peng, I recently submitted a proposal to ESA's **Quantum Technology for Space Exploration** call, investigating **endohedral fullerenes** as atomic clocks for space missions. Such clocks are unlikely to provide the highest absolute precision, but may provide advantages in size, weight, power and robustness. Our proposal passed the first round, and I attended the corresponding ESA workshop in July. This is precisely the type of comparison I want my sensing programme to enable: identifying the **architecture** with the strongest **competitive advantage under the relevant constraints**.

3.3: Relationship between quantum sensing and other quantum technologies

The same emphasis on realistic control also connects quantum sensing to the other major branches of quantum technology. Quantum **computers** and quantum **sensors** appear, at first, to face almost opposite challenges: computers must be protected from external disturbances, while sensors must respond strongly to them. In both cases, however, the central problem is to control which interactions are allowed to affect the quantum state and which should be suppressed. Techniques developed for quantum **computing** can therefore become useful for **sensing**. Similarly, techniques in quantum **communication** can enable **distributed quantum sensors** through entanglement sharing.

I am already exploring this connection through a consortium led by Prof. Hjorth-Jensen at UiO, which recently submitted an application to QuNorth's Quantum Flagship Challenge for early access to the Danish quantum computer **Magne**. The project aims to test quantum-computing methods that are also relevant for optimizing future quantum-sensor operations. In the longer term, I want to develop this into a broader programme on understanding the interplay of quantum sensing with **quantum computing and communication**. This also connects closely to the new national quantum centres.

To summarize, my planned future research is cumulative, following a natural progression. My work on topology asks which structures of a quantum state are globally protected. Quantum geometry asks how those structures constrain the local response of the state. Quantum sensing asks whether that response can provide enhanced information. Finally, quantum control lets us understand how the response can be **controlled and read out in actual devices**. I have already started work in each of these directions.

Research trajectory and synergies at UiO

The planned research directions compose my **research programme**, as summarized in the table.

In the **near term** (1-2 years) I plan to establish the foundations of this programme by extending the topology-geometry framework, study robustness of topological phases, develop the role of orbital embedding and complete high-throughput searches with AutoEuler. I also plan to extend my QFI measurement proposal toward realistic multiband systems, and extend my exploration of critical sensing. In parallel, the Eeroq and EQSITE work will establish how noise and experimental constraints modify theoretically optimal sensing protocols.

Research direction	Near term (1-2 years)	Medium term (3-5 years)	Long term (6-10 years)
1. Topology → Geometry			
1.1: Decompose geometry			
1.2: Entanglement			
1.3: Robustness			
1.4: Community tools			
2. Geometry → Sensing			
2.1: Measuring QFI			
2.2: Critical sensing			
3. Sensing → Control			
3.1 Emulating sensors			
3.2 Finding niches			
3.3: Relation to other technologies			

In the **medium term** (3-5 years), these strands should increasingly converge. I will study entanglement in

both materials and hardware architectures, within a common QFI and control framework, allowing me to identify which quantum resources remain valuable once realistic constraints are imposed.

In the **long term** (6-10 years), I aim to build a research group spanning the complete chain from **topological quantum matter to quantum sensing and control**. Particular targets include critical sensing at topological transitions, emulating sensors and the relationship between sensing and other quantum technologies. The unifying question will remain: **which structures of a quantum state are robust, which determine useful response, and how can that response be controlled and exploited?**

UiO provides an unusually strong environment in which to pursue this programme, as can be seen from my established collaborations summarized in the

	People	Type of collaboration	Shared interest
Physics			
SMN	Prof. Vines, Assoc Prof. Bathen	NRC Proposal, MSc. project proposals	Quantum materials, quantum sensing, DFT
CCSE	Prof. Hjorth-Jensen	Research project, research proposal, co-supervision, book writing	Quantum sensing, numerical methods
Theory	Prof. Massel	Research project, book writing	Quantum sensing, numerical methods
Condensed Matter	Prof. Bergli	Co-supervision, research project	Non-hermitian physics, Floquet engineering
Chemistry			
Hylleraas Center	Dr. Bore	Research project	DFT, AI in Science
Inorganic Chemistry	Assoc. Prof. Sønsteby	Guest lecturing course	Topological materials
Informatics			
Cybersecurity	Prof. Jøsang	Member of UiO Cyber interest group	Quantum technology in cybersecurity

table. My work in condensed-matter theory and materials modelling naturally connects theoretical quantum research in the Department of Physics with the experimental and computational activities of the **Centre for Materials Science and Nanotechnology (SMN)**. I already contribute to this connection through FYS-MENA4111, which I lecture and which attracts students from physics, chemistry and materials science, as well as through joint proposals, prospective co-supervised MSc projects and regular interaction with SMN.

The progression toward quantum sensing creates further connections to **DSQS**, while my work on electronic structure, geometry and entanglement overlaps naturally with the **Hylleraas Centre**. Sensor optimization and emulation create links to the Norwegian Computing Centre, while my longer-term interest in quantum protocols and control provides natural points of contact with **QOMBINE** and **QSTAR**.

At the same time, I will bring a distinct new research direction to UiO: the theory of **topological and geometrical quantum materials**. This is a major international field and one that I have found to be highly attractive to top students; both of my graduated research students are Aker Scholars and have prepared, or are preparing, scientific publications based on their work. As my programme is grounded in fundamental condensed-matter physics, it also has scientific relevance well beyond current quantum-technology priorities. I would seek to strengthen condensed-matter theory at UiO, including through future teaching in basic physics, modern condensed-matter theory and computational material physics.

My existing collaborations provide a strong basis for this programme. Internationally, I work with or am developing collaborations with groups in **Cambridge, Zurich, Singapore** and **Dresden**. In Norway, I am developing links across the emerging quantum-technology ecosystem, including with DNV, the Norwegian Computing Centre, the Norwegian Meteorological Institute and several universities and research institutes.

Furthermore, I have **already attracted funding** through my MSCA co-funded fellowship, YoungCAS grant, Aker Scholarship and support from the S. G. Sønneland Foundation. In addition to the submitted EQSITE proposal, I am preparing an NRC Young Research Talent application centred on quantum geometry and entanglement and **plan to apply** for an **ERC Starting Grant in 2027**. My aim is to use these opportunities to progressively build an independent group around my topology–geometry–sensing–control programme.

Summary

My research began with investigating **topology**. Work on real materials then led to the question: **what physical consequences does topology lead to?** Quantum geometry provides a natural answer by connecting global topology to the local response of a quantum state. That response, in turn, leads directly to **quantum sensing**, where sensitivity itself becomes a resource, and finally **quantum control**.

My future research will combine analytical theory, material simulations, computational tools and collaboration with experiments and industry, to understand **which measurable responses survive in real systems, and which can be exploited**. Over the coming decade, I aim to establish a research group at UiO that can follow this entire progression – from **topology, through geometry, to sensing and control**.

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